

Managing Expectations: Instruments vs. Targets

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Background

Monetary Policy Tools

- Forward guidance big component of MP post 2008
- Two different policy tools via communication available:

Instruments

Example

"Keep rates low until 2023"

Targets

Example

"Keep rates low until
unemployment <4%"

Two Agents in Setting

Rational Agents

Def. Know how govt. policies affect them *and others*

Example

I know how I and others will react to $\downarrow R$. $\implies$ I know output and other aggregates exactly

'Shallow-Reasoning' Agents

Def. Know how govt. policies affect them *but not others*

Example

I know $\downarrow R$ means I will spend more. But I don't know how others respond $\implies$ don't know output

Preview of Results

- Aim to Advise Fed on when to use **instrument-based** vs. **target-based** communication
- **Q.** Which tool is opt. for Fed who cares about efficient output, rates
- Agents fully believe Fed here... no commitment issues.
- **A.** Depends....

Rational Agents

Paper's Answer

Doesn't matter. Irrelevance result.
If Fed tells agent either policy rate or its target, agent know other tool.

Shallow Agents

Paper's Answer

Depends on how much Δ rate vs. Δ output affects size of distortion relative to first best. You should communicate the one that causes smaller distortions.

Model

Setting

- Two agent game
- **Player 1:** Identical agents from a continuum of agents $i \in [0, 1]$
 - Each agent chooses private choice $k_i \in \mathbb{R}$: 'private consumption'
 - Induces aggregate variable K : 'aggregate consumption' that is their average

$$K := \int_0^1 k_i di$$

- **Player 2:** Policy-maker has quadratic preferences in gap of interest rate and output relative to targets. Communicates either
 - $\tau \in \mathbb{R}$: 'interest rate'
 - $Y \in \mathbb{R}$: 'aggregate output'
- **State:** θ exogenous and random 'productivity' that policy-maker must know for optimization. Policy-maker can condition choices on θ

Model

- Three equations characterize model
- Objective func of policy maker w/ quadratic pref. over opt. rates and output
- Link between rates and output (aggregate rule)
- Agent's best response to an announced policy (agent's rule)
- agg. rule + agent's rule act as implementability constraint on policy maker
- Equations should be treated as entirely ad-hoc for time being

Full Description of Game Structure/Timing

3 periods, sequential game

1. θ is realized. The policy maker: observes θ , chooses to announce $\tau = \hat{\tau}$ for instrument communication or chooses τ such that $Y = \hat{Y}$ is achieved.
 - Fully commits to announcement
 - Min. objective: $-(\chi(Y - \theta)^2 + (1 - \chi)(\tau - \theta)^2)$, $\chi \in (0, 1)$, given agent BR
2. Each agent i hears announcement $\hat{Y}$ or $\hat{\tau}$, forms $\mathbb{E}_i[\cdot]$, chooses k_i following given (best-response) rule.
 - Agents symmetric, so $k_i = k$, $\implies k_i = K$
 - Agents believe policymaker: if $\hat{\tau}$ announced $\mathbb{E}_i[\tau \mid \hat{\tau}] = \hat{\tau}$, same for Y
 - Rational and shallow reasoning agents only differ in how $\mathbb{E}_i$ is formed.
3. K is computed; value of other instrument computed via:

$$Y = (1 - \alpha)\tau + \alpha K$$

1. Aggregate rule known to agents and policy maker (two formulations):

$$Y = (1 - \alpha)\tau + \alpha K, \quad \alpha \in (0, 1)$$
$$\Leftrightarrow \tau = \frac{\alpha}{1 - \alpha}Y - \frac{\alpha}{1 - \alpha}K$$

Definitions:

- Y : Aggregate Output
- τ : Policy rate
- K : average consumption/($= k_i$)consumption of single symmetric agent
- $\alpha \in (0, 1)$, $< M$: controls how much of $\Delta\tau$ is indirect via agent's choices vs. direct/mechanical effect .
- M is maximum value to avoid multiplicity of equilibria analogous to Taylor rule mult. (Atkeson, Chari, Kehoe 2010)

2. Agents' Rule

$$k_i = \underbrace{(1 - \gamma) \mathbb{E}_i[\tau]}_{\text{'PE'}} + \underbrace{\gamma \mathbb{E}_i[Y]}_{\text{'GE/Agg. Effect'}}, \quad \gamma \in (0, 1)$$

Definitions:

- Y : Aggregate Output
- τ : Policy rate
- k_i : chosen consumption of single symmetric agent
- γ : strength of GE feedback relative to PE effect. 'GE' : change in output and inflation--error free phillips curve, so linear combo
- $\mathbb{E}_i[\cdot]$: agents subjective expectations on aggregates

Model

- Substituting aggregate rules into Y or τ in agent's policy rule, we get:

$$\begin{aligned}k_i &= (1 - \gamma)\mathbb{E}_i[\tau] + \gamma\mathbb{E}_i[(1 - \alpha)\tau + \alpha K] \\ \Leftrightarrow k_i &= (1 - \gamma)\mathbb{E}_i\left[\frac{\alpha}{1 - \alpha}Y - \frac{\alpha}{1 - \alpha}K\right] + \gamma\mathbb{E}_i[Y]\end{aligned}$$

2.* Agents' Rule

$$\begin{aligned}k_i &= (1 - \alpha\gamma)\mathbb{E}_i[\tau] + \alpha\gamma\mathbb{E}_i[K], \quad \gamma, \alpha \in (0, 1) \\ \Leftrightarrow k_i &= \left(1 + \frac{(1 - \gamma)\alpha}{1 - \alpha}\right)\mathbb{E}_i[Y] - \frac{(1 - \gamma)\alpha}{1 - \alpha}\mathbb{E}_i[K] \quad \forall i \in (0, 1)\end{aligned}$$

Model

- Because we had a linear relationship between τ and Y were able to rewrite freely into linear form (a linear func of a linear func is again linear.)
- Defining:
 - $\delta_\tau := \alpha\gamma \in (0,1)$ intuition: $r \rightarrow Y \rightarrow k_i$
 - $\delta_Y := -\frac{(1-\gamma)\alpha}{1-\alpha} \in (-1,0)$
- We get simplified forms:

$$k_i = (1 - \delta_\tau)\mathbb{E}_i[\tau] + \delta_\tau\mathbb{E}_i[K]$$

$$k_i = (1 - \delta_Y)\mathbb{E}_i[Y] + \delta_Y\mathbb{E}_i[K]$$

- Assume the tool the policymaker communicates, agent knows.

Instrument Communication

$$k_i = (1 - \delta_\tau)\tau + \delta_\tau\mathbb{E}_i[K] = \lambda k_i$$

Target Communication

$$k_i = (1 - \delta_Y)Y + \delta_Y\mathbb{E}_i[K]$$

Policy Maker's Problem

- Define $X \in \{Y, \tau\}$ choice of communication
- From linear BR to communication policy (last slide):

$$k_i = (1 - \delta_X)X + \delta_X \mathbb{E}_i[K], \quad X \in \{\tau, Y\}$$

- Two reminders:
 - Agents symmetric $\implies K = k_i$
 - Assumed aggregate rule: $Y = (1 - \alpha)\tau + \alpha K$
- Plugging in linear BR k_i for K in agg. equation:
$$Y = (1 - \alpha)\tau + \alpha((1 - \delta_X)\hat{X} + \delta_X \mathbb{E}_i[K \mid \hat{X}])$$
- With (linear) rule for $\mathbb{E}_i[\cdot]$ Y announced $\implies \tau$ pinned down and vice-versa

Policy Maker's Problem

- Policy maker internalizes agents' response when making choice
- By assumption can only choose to communicate Y or τ , not combo
- By assumption, policy maker faces no tradeoff for first-best outcome:

$$Y^*(\theta) = \tau^*(\theta) = \theta$$

3. Policy Maker's Objective

$$\min_{(Y, \tau)} \chi(Y - \underbrace{\theta}_{Y^*(\theta)})^2 + (1 - \chi)(\tau - \underbrace{\theta}_{\tau^*(\theta)})^2, \quad \chi \in (0, 1)$$

s.t. (Y, τ) satisfy:

$$Y = (1 - \alpha)\tau + \alpha((1 - \delta_\tau)\tau + \delta_\tau \mathbb{E}_i[K \mid \tau]) \quad (\text{IC from Instr. Comm.})$$

OR

$$Y = (1 - \alpha)\tau + \alpha((1 - \delta_Y)Y + \delta_Y \mathbb{E}_i[K \mid Y]) \quad (\text{IC from Tar. Comm.})$$

Summary of Results

Rational Agents

- **Result:** W/ rational expectations + assumps. no diff. in policy tools
- Agents understand choices of others, $k_j = k_i$ so that $\mathbb{E}_i[K] = K \quad \forall i$
- As a reminder, best response for agents with:

Instrument communication (τ):

$$\underbrace{k_i}_{=K} = (1 - \alpha\gamma)\tau + \alpha\gamma \underbrace{\mathbb{E}_i[K]}_{=K}$$
$$\implies K = \tau$$

From $Y = \alpha K + (1 - \alpha)\tau$:

$$\implies K = Y = \tau$$

- First best achievable with $Y = \tau = \theta$

Target communication (Y):

$$\underbrace{k_i}_{=K} = \left(\frac{(1 - \gamma)\alpha}{1 - \alpha} + 1 \right) Y - \frac{\alpha(1 - \gamma)}{1 - \alpha} \underbrace{\mathbb{E}_i[K]}_{=K}$$
$$\implies K = Y$$

From $Y = \alpha K + (1 - \alpha)\tau$:

$$\implies K = Y = \tau$$

Shallow Reasoning Agents

- Assume: each agent believes $1 - \lambda$ agents don't pay attention to policy msg. λ trust policy maker fully, $\lambda \in (0, 1)$.

$$\mathbb{E}_j[X] = \begin{cases} X & \text{w.p. } \lambda \\ 0 & \text{w.p. } 1 - \lambda \end{cases}, \quad X \in \{Y, \tau\}$$

- Can show $\implies \mathbb{E}_i[K] = \lambda K$
- Intuition: if λ fraction of agents notice X , BR of λ agents is $k_i = K$ symmetric, rest is 0 $\implies \mathbb{E}_i[K] = \int \mathbb{E}[k_j]dj = \lambda k_i = \lambda K$
- In reality everyone responds $\implies k_i = K$
- Produces a multiplier effect

Shallow Reasoning Agents

- Key insight: first best has multiplier 1, no dist. Want multiplier closest to 1
- Plugging expectations and $k_i = K$ into linear BR:

$$K = (1 - \delta_X)X + \delta_X \lambda K$$
$$\implies K = \underbrace{\frac{1 - \delta_X}{1 - \lambda \delta_X}}_{:= \kappa_X} X$$

- Reminder: $-\frac{(1-\gamma)\alpha}{1-\alpha} = \delta_Y \leq 0 \leq \delta_\tau = (1 - \gamma\alpha)$ under param restriction:
- This implies multipliers:

$$\kappa_\tau \leq 1 \leq \kappa_Y$$

- What really matters is size of distortion: $|\kappa_\tau - 1| \lesssim |\kappa_Y - 1|$. Want to pick instrument with smaller distortion (since we can only pick one.)
- Linearity $\implies K$ distortion $\propto$ distortion in τ, Y , so κ_X 'distortion' char. welfare distortion. Another way to say is $\kappa_X \propto$ slope μ , where $Y = \mu\tau$

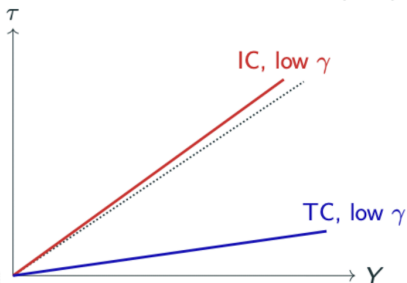
Main Result + Intuition

- Main result of paper: if GE effect size
 - $\gamma > \bar{\gamma}$, prefer target communication (Y),
 - if $\gamma < \bar{\gamma}$ prefer instrument communication (τ)
- Relative size of distortions controlled by γ via 2 intermediate linear eqs.
- Linearity $\implies$ cutoff value $\bar{\gamma}$ where policy tool with largest dist. flips
- Prefer τ when $1 - \kappa_\tau < \kappa_Y - 1$ for fixed λ
- Implies $\delta_\tau - 0 < \bar{\delta} < 0 - \delta_Y \bar{\delta}$ where κ distortion equal
- Holding α fixed, $\implies \gamma < \bar{\gamma}$ by δ_X positive linear in γ
- γ controls 'sensitivity' of distortion to output communication, for low γ , large change in $Y \implies \tau$ small change, small effect on distortion.

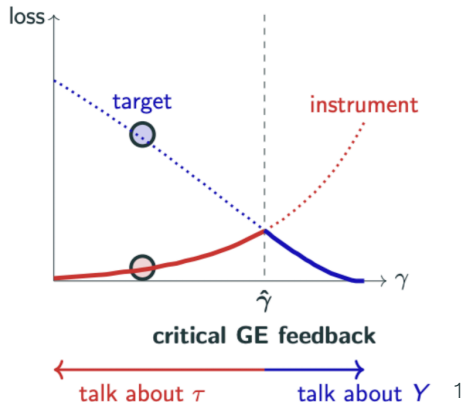
Main Result Visualized (Optimal Policy Minimizes Distortion Among Feasible (τ, Y))

$\kappa_X \propto 1/\text{slopes in left figure}$

Implementable combinations of (τ, Y)



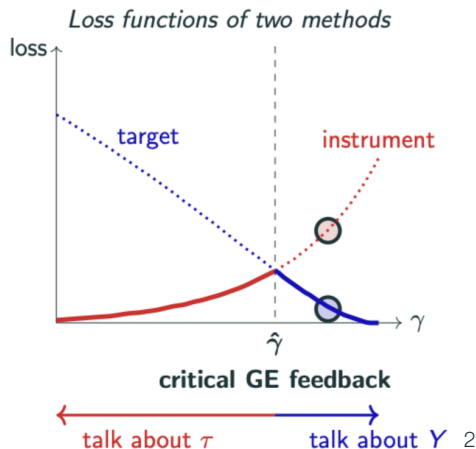
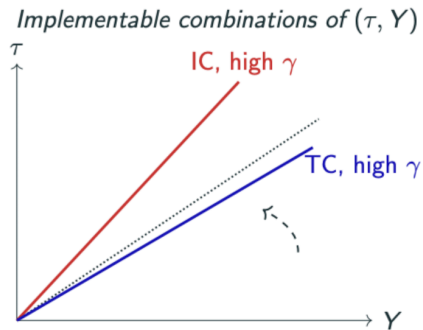
Loss functions of two methods



¹Graph courtesy of Angeletos: <https://economics.mit.edu/files/22014>

Main Result Visualized (Optimal Policy Minimizes Distortion Among Feasible (τ, Y))

$\kappa_X \propto 1/\text{slopes in left figure}$



²Graph courtesy of Angeletos: <https://economics.mit.edu/files/22014>

Connection to Monetary Policy

- Authors argue that setting chosen 'looks like' NK economy at ZLB $\implies$ guidance for policy makers based on size of GE effects
- Present an 'ad-hoc' New Keynesian economy whose log-linearization nearly coincides with their model under special conditions
- Authors suggest γ equivalent increasing with MPC and price flexibility

NK Economy

Results-NK Economy

Conclusion

- In a simple, analytically clear model, paper describes conditions when target communication more effective than instrument communication
- With rational agents in their model, no difference
- With 'shallow-reasoning' agents, comes down to which is less distortionary relative to RE benchmark, simplifies to cutoff rule in γ
- Model simple, enables authors' (kth order logic style reasoning)
- Arguably may not be microfounded sufficiently/assumptions not distanced from results
- Linearity, no combo of rules, style of expectations no other distortions crucial to result
- Authors leave to future work to relax some of these assumptions

Appendix

A NK Microfoundation for Model

- Consumer objective:

$$\begin{aligned}\max_{B_{it}} V_{it} &= \mathbb{E}_{it} \left[\left(\log C_{it} - \frac{1}{2} N_{it}^2 \right) + \beta V_{it+1} \right] \\ s.t. & C_{it} + B_{it} = R_{t-1} \frac{P_{t-1}}{P_t} B_{it-1} + Y_{it}\end{aligned}$$

- $\beta_t := \exp(-\hat{\rho} - \rho_t)$, $\hat{\rho}$ SS discounting
- Technology
 - Final goods: $Y_t = e^{\theta_t} F((X_{jt})_{j \in [0,1]})$, F CES aggregator.
 - Intermediates $P_{jt} = P_t$ and $X_{jt} = X_t = N_t$
- 'Ad-hoc' backwards looking phillips curve:

$$\frac{P_t}{P_{t-1}} = \left(\frac{Y_{t-1}}{Y_{t-1}^*} \right)^\xi$$

- Shocks: discount factor ρ_t , TFP θ_t

[Return to MP Connection](#)

Results from NK Model

When

- this model is log-linearized around SS,
- a particular sequence of shocks are Fed over three periods,
- agents are rational or shallow reasoning
- the Fed faces a lower bound on nominal constraints that prevent it from implementing first-best policy

we almost recover the specification of the original problem when we also impose a nominal lower constraint on R

- We will also require the liquidity trap 'weakly bind' in which case we recover the original model.